Auditory Superstitious-Perception: results walkthrough

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1 The experiment

26 college students completed two blocks of ∼150 trials each. On every trial they heard a short noisy audio clip and reported whether it was the "target" or the "distractor". The underlying target and distractor clips are essentially noise; chance performance is 0.50.

The two block types are `full_sentence` (the target is replayed at the start of each trial) and `imagined_sentence` (subjects must hold the target in memory). Subjects were recruited at Oberlin ($n = 16$) and UChicago ($n = 10$). Every comparison is reported three ways: each site alone and pooled ("Unified").

2 Brief jargon primer

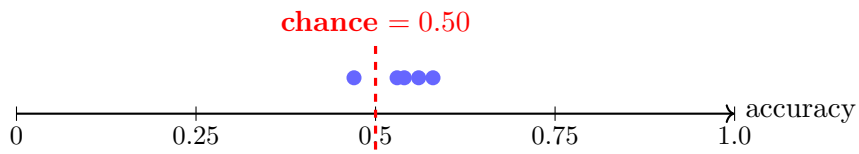
Skip if you already know what a p-value, Cohen's d, and cosine similarity are.

2.1 Accuracy and chance

Each subject's score is fraction correct, between 0 and 1. If the noise is truly undiscriminable, scores cluster around 0.50.

The intuition: imagine flipping a fair coin 150 times and calling each flip. You will get close to 75 right but almost never exactly 75. By chance alone you might land at 80/150 (0.533) or 70/150 (0.467). That random spread around 0.50 is what a single subject looks like under the null. The question for each group is whether the cloud of subject means sits noticeably to the right of 0.50 *more often than that random spread would explain.*

With 150 trials per subject, the standard deviation of an individual subject's accuracy under the null is roughly $\sqrt{0.25/150} \approx 0.041$. So a single subject scoring 0.54 is unremarkable; a group of 16 subjects *averaging* 0.526 is much harder for chance to produce, because the group mean's standard error shrinks by $\sqrt{16}$.



2.2 The p-value

Probability of a result at least as extreme as the observed one, under the null hypothesis. Smaller p = more surprising under the null. The conventional cutoff is $p < 0.05$.

The intuition: pretend the null is true (subjects are guessing). Now imagine re-running the whole experiment a thousand times in that imaginary world. The p-value is the fraction of those imaginary replications that would produce a group mean at least as far from chance as what we actually observed. $p = 0.05$ means *1 in 20 imaginary null-world experiments would look this extreme by accident.* $p = 0.001$ means 1 in 1000.

A p-value is *not* the probability that the null is true, and it is *not* the probability that the effect is real. It is a statement about how often pure noise would mimic the observation.

When we run many tests (e.g. 6 cells in Section A, 48 correlations in Section F) some will land below 0.05 by accident. Bonferroni correction divides $\alpha = 0.05$ by the number of tests; so 6 tests $\Rightarrow$ require $p < 0.0083$, 48 tests $\Rightarrow$ require $p < 0.001$.

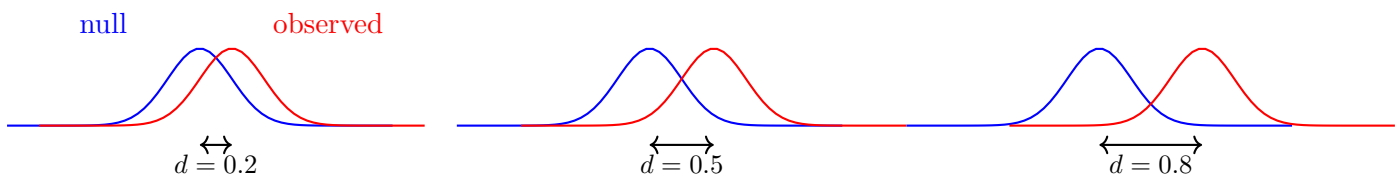
2.3 Cohen's d

Effect size in units of pooled standard deviation. Rough rule of thumb: $|d| \approx 0.2$ small, 0.5 medium, 0.8 large.

The intuition: p-values conflate two things — how big the effect is, and how many subjects you ran. Run enough subjects and a tiny effect will clear $p < 0.05$; run too few and a large effect will miss it. Cohen's d strips out sample size and tells you the size of the gap between the observed mean and the null in units of how spread out the subjects are.

A concrete way to read it: $d = 1.0$ means the observed mean is one full standard deviation away from the null value. If subjects' scores formed a bell curve centered on chance, an observed group mean at $d = 1.0$ would sit out near the edge of that bell, in the region where only $\sim 16\%$ of single

subjects naturally fall. The picture below shows three null/observed pairs at $d = 0.2, 0.5, 0.8$ — notice how much the two distributions still overlap even at $d = 0.8$.

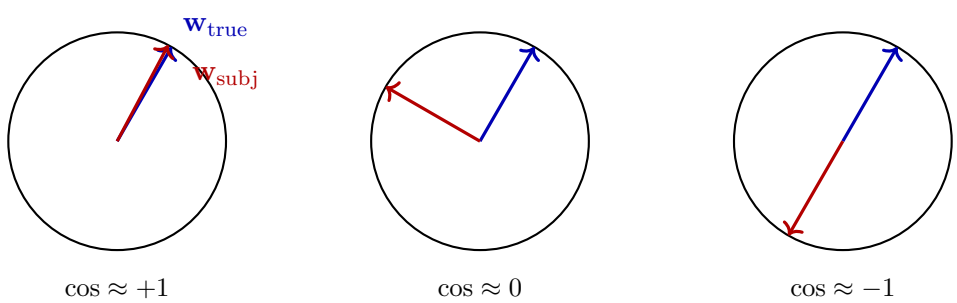


2.4 Cosine similarity

A waveform or per-subject classification image is a long list of numbers, i.e. a vector. The cosine of the angle between two vectors is $+1$ if identical direction, 0 if orthogonal, -1 if opposite. We use it to compare a subject's classification image to the true target waveform.

The intuition: a subject's *classification image* is built by averaging the audio waveforms of all the trials they called "target" and subtracting the average of trials they called "distractor". If they were guessing, this difference is just averaged noise and points in a random direction. If they were actually picking up on the target, this difference should look something like the real target waveform — so the angle between their classification image and the true target should be small, and the cosine close to $+1$.

Why cosine and not raw dot product: cosine ignores how loud or contrasty the subject's classification image happens to be (its length) and only asks whether it points in the right direction. Two subjects with very different overall response biases can both have $\cos = +0.5$ if their classification images both *shape-match* the target.



2.5 d' (signal-detection sensitivity)

d' separates "can the subject tell target from distractor" from "does the subject lean toward saying yes or no". Two subjects can both score 0.55 accuracy, but one might be balanced (hits and correct rejections both around 0.55) while the other is biased (says "target" on almost every trial, racking up hits but also false alarms). d' rewards the first and not the second.

Concretely, $d' = z(\text{hit rate}) - z(\text{false-alarm rate})$. Null value is 0. As a sanity check, an accuracy of 0.526 with no bias corresponds to $d' \approx 0.13$, matching the Oberlin full-sentence row in Section E.

2.6 $\beta_{\textbf{truetarget}}$ (regression slope on the true target)

A logistic regression of the subject's per-trial response on the trial-by-trial dot product between the stimulus waveform and the true target waveform. A positive slope means *the more the trial's audio*

resembled the true target, the more likely the subject was to call it "target". Null value is 0. This is a direct test of target-driven responding, with units of log-odds per unit of target-similarity.

3 Results

3.1 Section A: Accuracy vs chance

Section A (`A_accuracy_vs_chance.csv`)

block	group	n	mean	95% CI	Wilcoxon p	d
full_sentence	UChicago	10	0.504	$[0.479, 0.530]$	0.869	+0.09
full_sentence	Oberlin	16	0.526	$[0.513, 0.538]$	0.0037	+0.99
full_sentence	Unified	26	0.518	$[0.504, 0.530]$	0.019	+0.50
imagined_sentence	UChicago	10	0.490	$[0.464, 0.515]$	0.611	−0.23
imagined_sentence	Oberlin	16	0.510	$[0.495, 0.526]$	0.277	+0.31
imagined_sentence	Unified	26	0.502	$[0.488, 0.517]$	0.517	+0.06

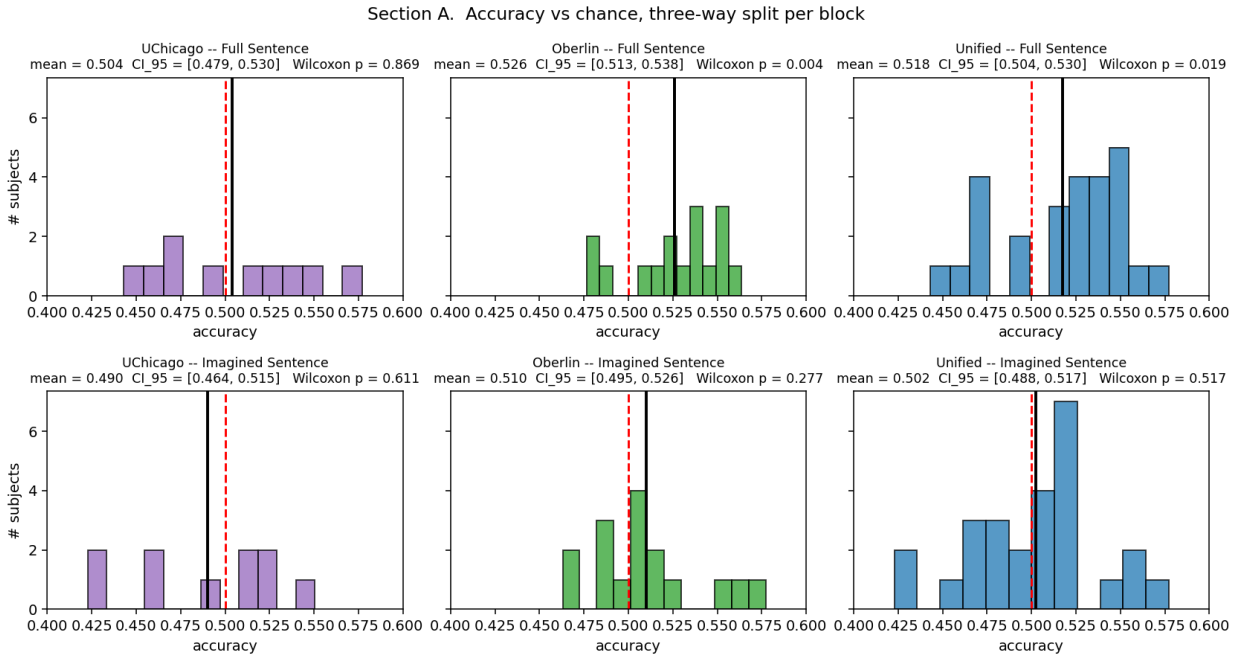


Figure 1: Per-subject accuracy by (block, group). Red dashed = chance (0.50); black solid = group mean. Title shows the bootstrap CI on the mean and the Wilcoxon p vs. chance.

Bonferroni-corrected threshold across the six tests is $\alpha = 0.05/6 \approx 0.0083$. Oberlin full-sentence ($p = 0.0037$) clears it; the Unified full-sentence cell ($p = 0.019$) does not.

3.2 Section B: Full vs Imagined (paired)

Section B (`B_full_vs_imagined.csv`)

group	n	mean(F)	mean(I)	mean diff	paired Wilcoxon p	perm p	d
UChicago	10	0.504	0.490	+0.0141	0.543	0.508	+0.22
Oberlin	16	0.526	0.510	+0.0159	0.209	0.198	+0.35
Unified	26	0.518	0.502	+0.0152	0.157	0.158	+0.29

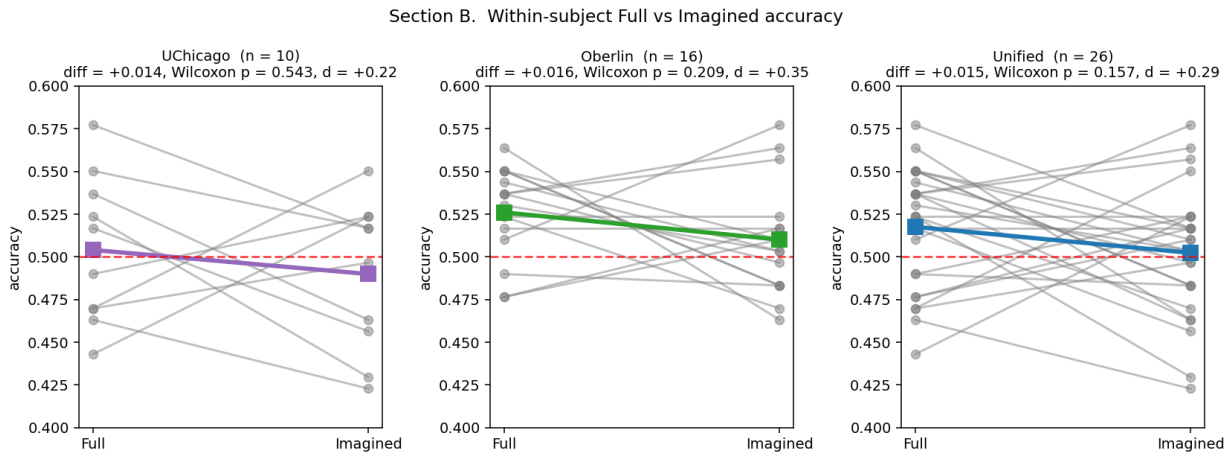


Figure 2: Per-subject lines connecting Full to Imagined accuracy in each group; coloured heavy line is the group mean.

The mean difference is +0.0152 pooled and positive in both sites. No paired test reaches $p < 0.05$.

3.3 Section C: Site comparison

Section C (`C_site_comparison.csv`)

block	n_O	n_U	mean$_O$	mean$_U$	diff	MWU p	Welch p	d
full_sentence	16	10	0.526	0.504	+0.0220	0.145	0.173	+0.65
imagined_sentence	16	10	0.510	0.490	+0.0201	0.475	0.231	+0.54

Oberlin is numerically higher than UChicago in both blocks by ≈ 0.02. Neither between-site comparison reaches $p < 0.05$.

3.4 Section D: Block order

Section D selected rows (`D_block_order.csv`)

block	group	n_{B1}	n_{B2}	mean(B1)	mean(B2)	MWU p	d
full_sentence	Oberlin	5	11	0.519	0.529	0.493	+0.35
imagined_sentence	Oberlin	11	5	0.500	0.532	0.078	+1.03

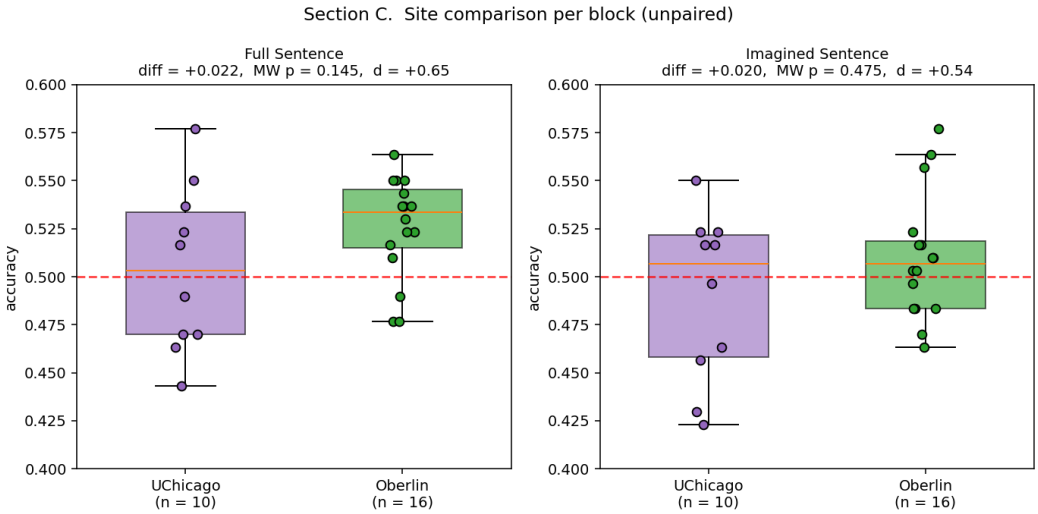


Figure 3: Per-block boxplots, Oberlin vs. UChicago, with overlaid jittered subject points.

In the Oberlin imagined block, subjects who completed it second ($n_{B2} = 5$) averaged 0.532, vs 0.500 for those who completed it first ($n_{B1} = 11$); MWU $p = 0.078$, $d = +1.03$. All other block-order cells have $p > 0.4$.

3.5 Section E: Convergent metrics, full-sentence block

Section E (`E_convergent_full_sentence.csv`)

group	metric	mean	95% CI	Wilcoxon p	d
Oberlin	d′	+0.137	[+0.069, +0.200]	0.0013	+0.99
Oberlin	accuracy	+0.526	[+0.513, +0.538]	0.0037	+0.99
Oberlin	cos(CI, true)	+0.056	[+0.029, +0.083]	0.0013	+1.00
Oberlin	beta_truetarget	+0.232	[+0.118, +0.340]	0.0027	+0.99
UChicago	d′	+0.021	[−0.106, +0.153]	0.846	+0.09
UChicago	cos(CI, true)	−0.002	[−0.059, +0.056]	1.000	−0.02
UChicago	beta_truetarget	+0.042	[−0.159, +0.248]	0.846	+0.12
Unified	d′	+0.093	[+0.023, +0.159]	0.013	+0.51
Unified	cos(CI, true)	+0.034	[+0.004, +0.063]	0.025	+0.43
Unified	beta_truetarget	+0.159	[+0.047, +0.267]	0.016	+0.55

At Oberlin all four metrics yield $p \leq 0.0037$ and $d \approx 1.0$. At UChicago all four are non-significant with $|d| \leq 0.12$. Pooled, all four yield $p \in [0.013, 0.025]$ and $d \in [0.43, 0.55]$.

3.6 Section F: Questionnaire correlations

48 Spearman correlations (8 questionnaires × 2 blocks × 3 groups). Bonferroni threshold $p < 0.001$. None survive correction. Tables: `F_questionnaire_spearman.csv`; figures: `F_quest_vs_accuracy_full_sentence.` `F_quest_vs_accuracy_imagined_sentence.png`.

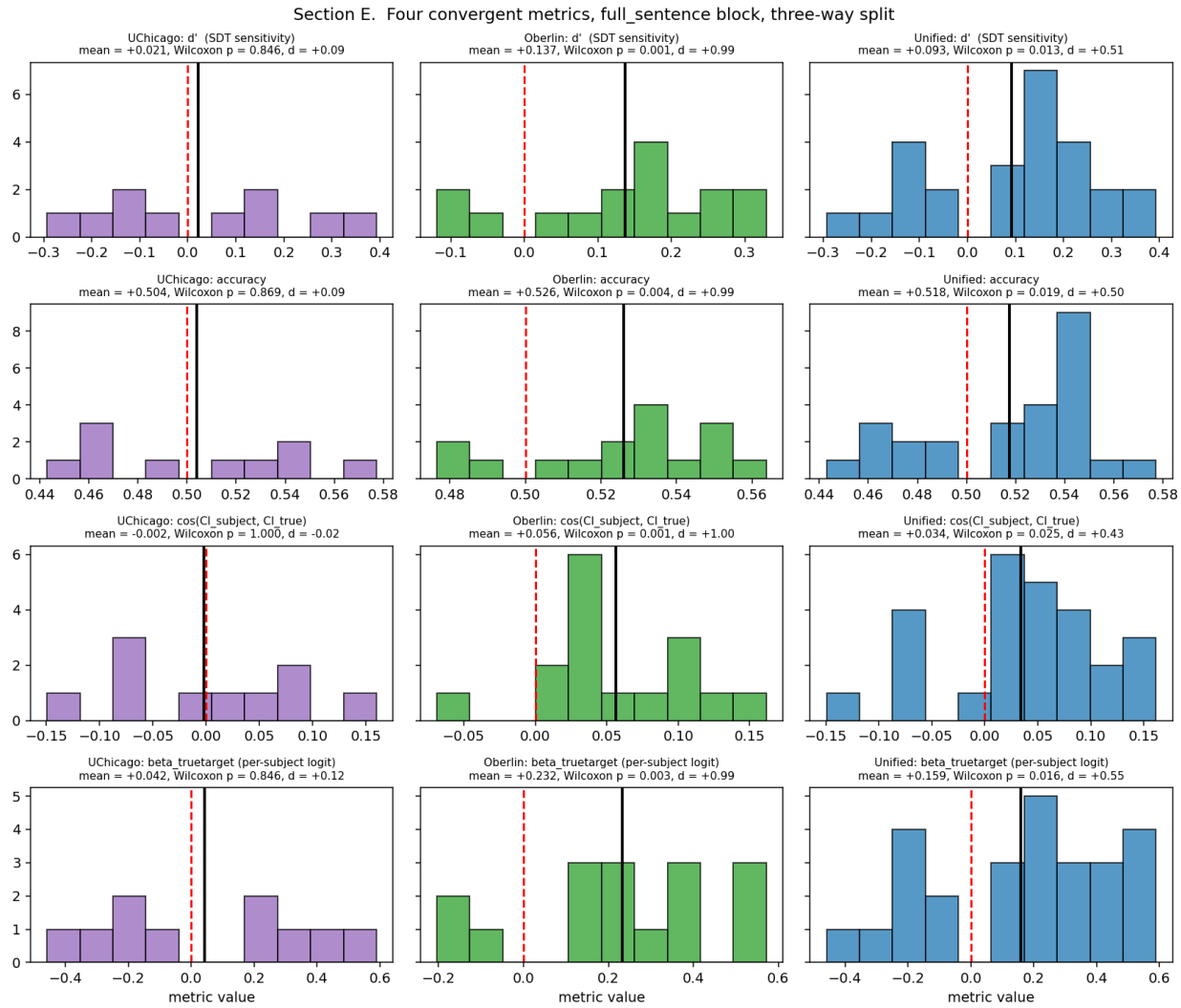


Figure 4: Each row is a metric, each column a group. Red dashed = null value (0 or 0.5); black solid = observed mean.

4 Further analyses available with the existing data

4.1 Trial-by-trial mixed-effects modelling

The current per-subject summaries collapse ∼150 trials per block into a single mean. A mixed-effects logistic regression on raw trials, with subject as a random effect, would use the within-subject trial variance directly.

4.2 Bayesian re-analysis of trending comparisons

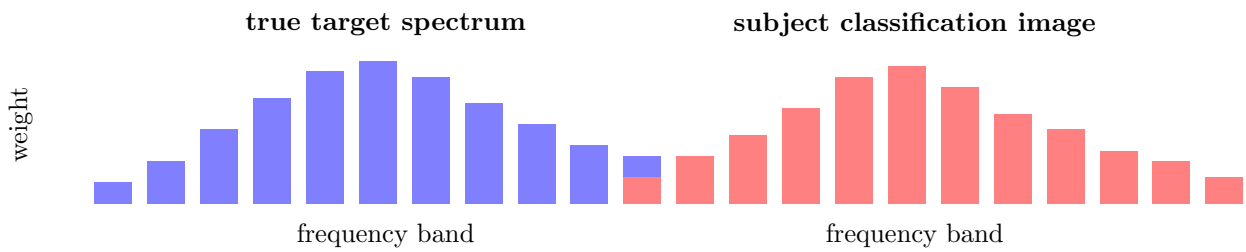
For comparisons with $0.05 < p < 0.20$ (Section B Full–Imagined paired, Section C site differences), Bayes factors quantify the evidence ratio between H_1 and H_0 rather than only rejecting H_0.

4.3 Within-subject correlations between convergent metrics

The four Section E metrics are tested independently. Computing the per-subject correlation matrix between d', accuracy, $\cos(\mathrm{CI}, \mathrm{true})$, and $\beta_{\mathrm{truetarget}}$ tests whether they are driven by a single latent perceptual ability.

4.4 Spectral classification image

A 32-band spectral analogue of the time-domain classification image: for each subject, fit per-band weights and compute the cosine with the true target's per-band spectrum.



4.5 Block-order follow-up in the Oberlin imagined block

Trial-resolved analysis of the $n_{B2} = 5$ vs $n_{B1} = 11$ split: e.g. within-block accuracy time course, and whether the $+0.032$ mean difference is concentrated in early or late trials of the block.

4.6 Stimulus-level analyses

Per-stimulus accuracy across the audio set, to test whether particular target/distractor pairs drive the Oberlin full-sentence effect. Building blocks for this exist in `stimuli-analyses/` (classification-image, spectral logistic, etc.).